

WASTE SORTING ASSISTANCE FOR RECYCLING

A PROJECT REPORT

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In fulfilment for the award of the degree

of

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***LDRP INSTITUTE OF TECHNOLOGY AND
RESEARCH
GANDHINAGAR***

CE-IT Department



CERTIFICATE

This is to certify that the Project Work entitled **“Waste Sorting Assistance for Recycling”** has been carried out by **Panchal Mansi Shashikant (23BEIT30080)** under my guidance in fulfilment of the degree of Bachelor of Engineering in <Name of the Department> Semester-6 of Kadi Sarva Vishwavidyalaya during the academic year 2025-2026.

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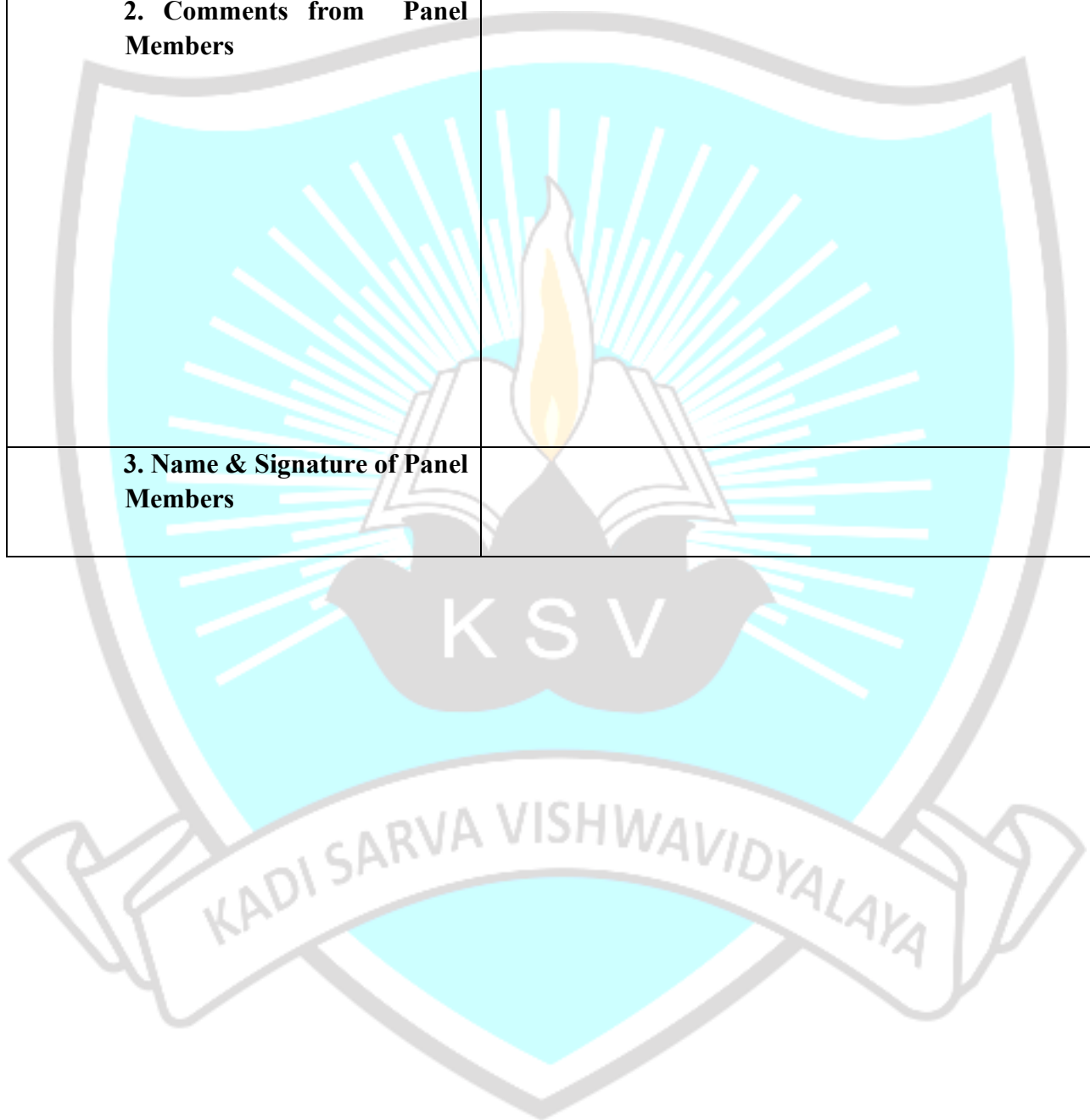
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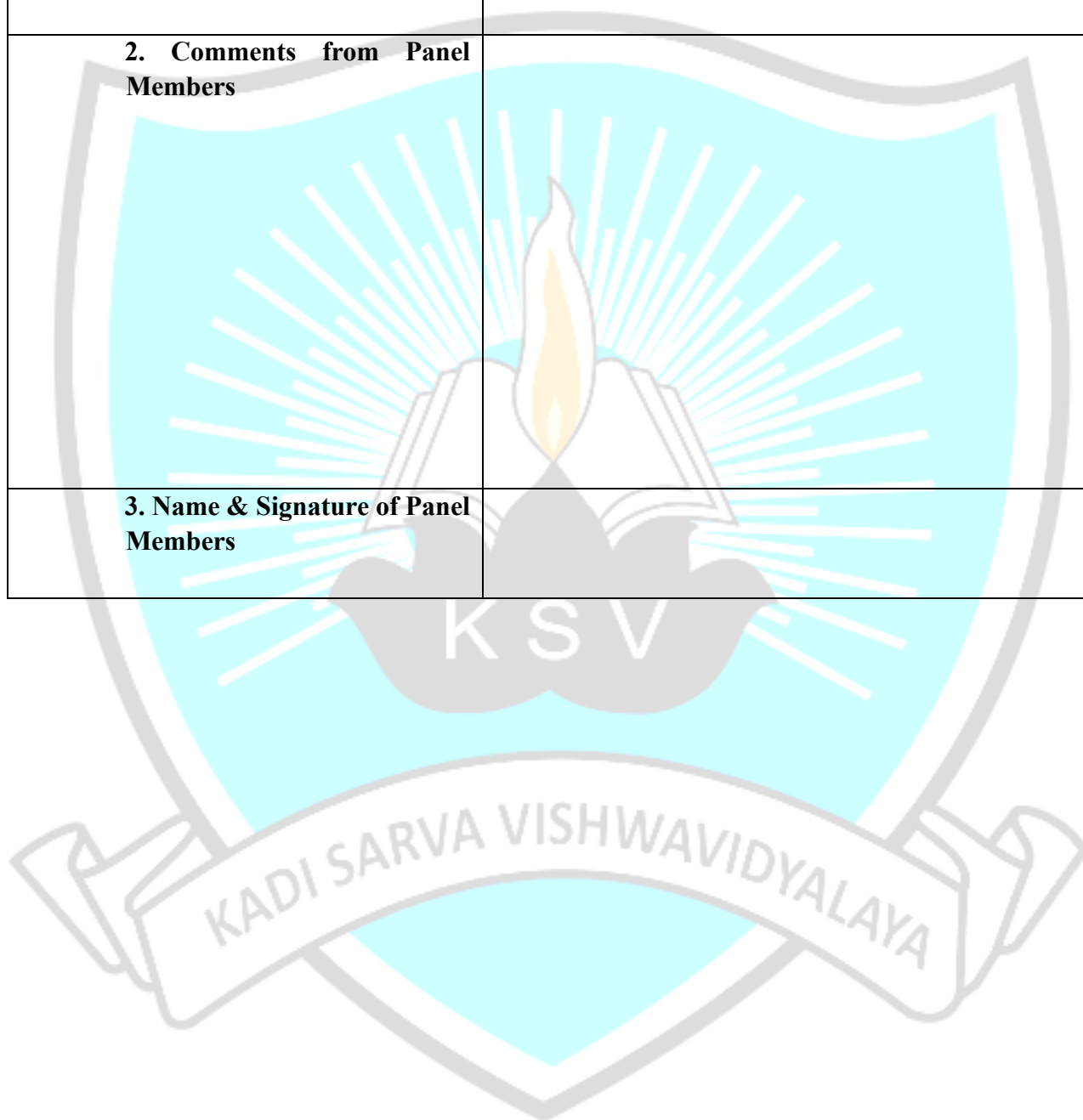
Presentation-I for Project-I

1. Name & Signature of Internal Guide	
2. Comments from Panel Members	
3. Name & Signature of Panel Members	



Presentation-II for Project-I

1. Name & Signature of Internal Guide	
2. Comments from Panel Members	
3. Name & Signature of Panel Members	



No.	Title of Unit
1	Introduction
2	Technology and Literature Review
3	System Requirements Study
4	System Diagrams
5	Data Dictionary
6	Result, Discussion and Conclusion
7	References

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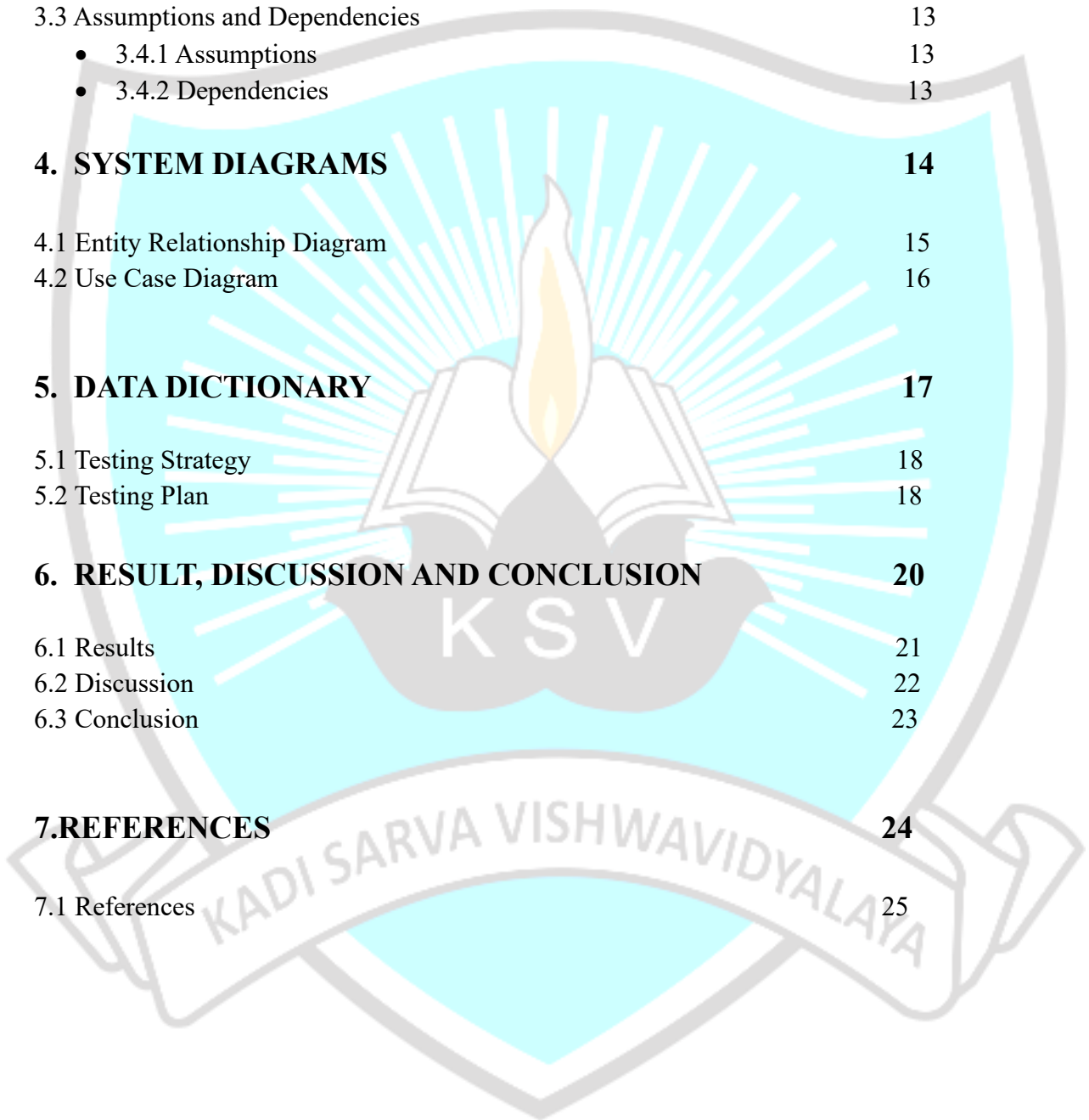
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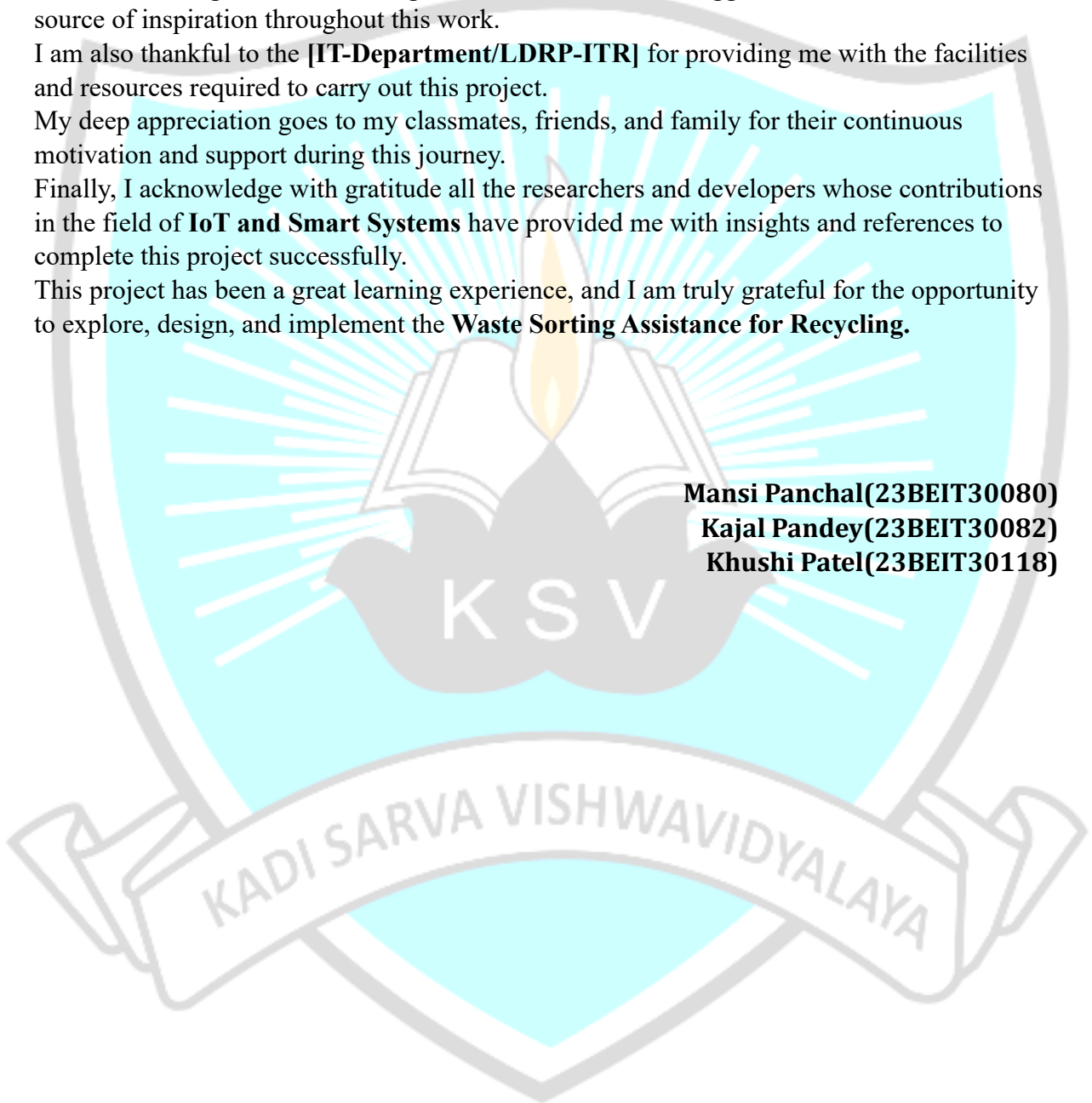
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ABSTRACT

The **Sortify** website is an innovative digital platform designed to **simplify the way users sort, manage, and dispose of waste efficiently**. This system integrates smart guidance to provide an interactive and user-friendly environment for identifying recyclable materials, reducing environmental pollution, and promoting sustainable habits across India.

The main goal of the system is to **digitize the traditional waste management process** by allowing users to access disposal guides anytime and anywhere through a centralized hub. It incorporates features such as waste categorization, local recycling centre mapping, and a **beginner-friendly interface** to make recycling easier for the everyday citizen. By focusing on real-life problems faced in the Indian context, Sortify aims to bridge the gap between household waste and formal recycling sectors.

The project emphasizes **automation, accessibility, and efficiency**, ensuring that users—including students, households, and small businesses—can seamlessly contribute to a cleaner environment. By implementing a straightforward design and smart categorization tools, Sortify bridges the gap between manual waste dumping and **efficient recycling in India**, making the journey toward a greener future more intuitive for everyone.

KADI SARVA VISHWAVIDYALAYA

1 INTRODUCTION

- **INTRODUCTION**
- **AIMS AND OBJECTIVE OF THE WORK**
- **BRIEF LITERATURE REVIEW**
- **PROBLEM DEFINITION**
- **PLAN OF THEIR WORK**



1.1 Introduction

Sortify is an integrated digital platform designed to automate, manage, and track the process of waste segregation and disposal. In the Indian context, the rapid growth of urban areas has led to significant challenges in waste handling, where the lack of proper sorting at the source often leads to **environmental pollution**. The primary goal of implementing Sortify is to enhance the efficiency of the recycling chain, improve the accuracy of waste categorization, and provide a superior, **beginner-friendly** experience for citizens. By replacing confusing or non-existent manual sorting habits with digital guidance, the system ensures that recyclable materials are easily diverted from landfills and managed through a central, organized platform.

The necessity for Sortify has grown significantly as Indian cities expand and the volume of daily waste increases. Modern households and businesses generate vast amounts of diverse waste, including plastics, paper, organic matter, and increasingly, electronic waste. An effective platform like Sortify provides a **robust database** to handle this complexity, offering easy-to-follow instructions, real-time recycling center tracking, and educational resources. This functionality is crucial for individuals who want to contribute to **efficient recycling in India** but may not know where to start. Essentially, the system serves as the digital backbone that connects the public with the recycling ecosystem.

Ultimately, a well-implemented Sortify platform delivers profound benefits to all stakeholders. For **users**, it simplifies the process of identifying waste types and finding nearby disposal points through an intuitive interface. For **recycling agencies**, it significantly improves the quality of collected materials, reducing the time spent on manual secondary sorting and allowing them to focus on core processing activities. In an era where sustainability is key, Sortify is not just a tool for waste; it is a vital platform for improving the quality of the environment and ensuring that Indian communities remain clean, green, and engaged in responsible living.

1.2 Aims and the objective of the work

Aim

The primary aim of **Sortify** is to develop a beginner-friendly digital platform that simplifies waste segregation at the source to combat **environmental pollution** and promote **efficient recycling in India**.

Objectives

To achieve the stated aim, the following objectives have been established:

- **To Simplify Waste Categorization:** Create an easy-to-use interface that helps users identify and sort waste into categories (Dry, Wet, E-waste, etc.) without needing prior technical knowledge.
- **To Bridge the Knowledge Gap:** Provide a centralized hub of information tailored to the Indian context, making the recycling process less intimidating for beginners.
- **To Reduce Environmental Pollution:** Minimize the amount of mixed waste reaching landfills by encouraging proper sorting at the household and organizational levels.
- **To Enhance Accessibility:** Ensure that disposal guides and recycling information are accessible anytime and anywhere through a responsive web platform.
- **To Promote Automation in Waste Management:** Implement smart search and categorization features that reduce the manual effort required to figure out "what goes where."
- **To Support Local Ecosystems:** Connect users with local recycling centers and collection points to ensure that sorted waste actually enters the formal recycling stream.

1.3 Literature Review

The study of waste management reveals that while India has robust recycling industries, the primary challenge remains at the "source"—the household level. This literature review explores why a digital intervention like **Sortify** is necessary to solve real-life environmental problems.

Current Challenges in Indian Waste Management

- **Lack of Source Segregation:** Research shows that a significant portion of urban waste in India is collected in a mixed state, which leads to high levels of **environmental pollution** and makes **efficient recycling** nearly impossible.
- **Information Overload vs. Action:** While many government awareness programs exist, citizens often find the specific rules for different materials (like multi-layered plastics vs. high-density plastics) confusing and difficult to follow.

- **The "Beginner" Barrier:** Most existing waste management systems are designed for industrial users or experts. There is a clear lack of **beginner-friendly** tools that guide an average citizen through the first steps of waste sorting.

The Role of Digital Solutions

- **Accessibility and Centralization:** Traditional methods of spreading awareness (flyers, TV ads) are often forgotten. A centralized digital hub ensures that sorting information is available **anytime and anywhere**, much like modern digital library systems.
- **Automation of Knowledge:** By implementing smart categorization, technology can take the guesswork out of recycling. Instead of a user researching "how to dispose of an old battery," a platform like Sortify provides an immediate, structured answer.
- **Bridging the Gap:** Studies suggest that when recycling is made "easier" and more "intuitive," participation rates increase by over 40%. Sortify aims to leverage this by focusing on a simple interface and clear data handling.

1.4 Problem Definition

The current state of waste management in India faces a critical crisis due to the lack of segregation at the source. While the country has a massive recycling network, it is crippled by the way waste is initially handled. The core problems identified are as follows:

1. Environmental Pollution through Mixed Waste

In most Indian households, dry, wet, and hazardous wastes are disposed of in a single bin. This mixture ends up in massive landfills where it undergoes anaerobic decomposition, releasing toxic leachates into the soil and greenhouse gases into the atmosphere, directly contributing to **environmental pollution**.

2. Inefficiency in the Recycling Chain

For **efficient recycling in India**, materials must be clean and separated. When recyclables (like paper or plastic) are contaminated by organic food waste, their value drops to zero. This makes the work of "Kabadiwalas" and formal recycling plants significantly harder and less productive.

3. Complexity and Lack of Guidance

The average citizen often wants to recycle but finds the process overwhelming. Existing rules are often technical and hard to find. There is a distinct lack of a **beginner-friendly** tool that tells a user exactly how to clean, sort, and where to give their specific waste items.

4. The "Last-Mile" Information Gap

Currently, there is no centralized, **digital platform** that connects an individual's household waste to the nearest authorized recycling center. This gap leads to "convenience dumping," where people throw away recyclables simply because they don't know a better alternative exists.

To ensure the successful development and implementation of **Sortify**, the project is divided into specific, manageable phases. This structured plan focuses on creating a **beginner-friendly** experience while ensuring technical robustness for **efficient recycling**.

Phase 1: Research and Data Collection

- **Gap Analysis:** Identifying the most common types of household waste in Indian urban areas that contribute to **environmental pollution**.
- **Data Sourcing:** Compiling a comprehensive database of local recycling centers, "Kabadiwala" networks, and government disposal guidelines.
- **User Persona Mapping:** Defining the needs of "beginner" recyclers to ensure the interface remains simple and non-intimidating.

Phase 2: System Design and Architecture

- **UI/UX Prototyping:** Designing a clean, intuitive interface that focuses on **accessibility**, using visual icons for different waste categories (Plastic, Organic, E-waste, etc.).
- **Database Schema:** Designing a **robust database** capable of handling categorization logic and location-based searches for recycling hubs.
- **Algorithm Development:** Creating the logic for the search functionality to provide instant disposal instructions anytime and anywhere.

Phase 3: Development and Implementation

- **Frontend Development:** Building the website using modern frameworks to ensure it is responsive across mobile and desktop devices.
- **Backend Integration:** Connecting the user interface to the central data hub to allow for real-time information retrieval.
- **Automation Features:** Implementing smart sorting guides that help users categorize their waste through a simple, step-by-step digital process.

Phase 4: Testing and Quality Assurance

- **Usability Testing:** Conducting trials with non-technical users to ensure the platform makes recycling truly "easier."
- **Data Validation:** Verifying that the recycling center locations and sorting advice are accurate and specific to the Indian context.
- **Performance Optimization:** Ensuring the platform is fast and reliable for daily household use.

Phase 5: Deployment and Outreach

- **Launch:** Deploying the **Sortify** website for public use.
- **Community Engagement:** Partnering with local organizations to encourage the adoption of the platform, aiming to bridge the gap between manual dumping and digital sorting.

2 TECHNOLOGY AND LITERATURE REVIEW

- TOOLS AND TECHNOLOGY
- PROJECT PLANNING



2.1 Tools and Technologies

The development of **Sortify** utilizes a modern web development stack. This combination of "Frontend," "Backend," and "Database" ensures the platform is fast, secure, and **beginner-friendly**.

1. Frontend: The User Experience

The frontend is the "face" of Sortify. It is designed to be accessible anytime and anywhere, ensuring even a first-time user can navigate the waste sorting process easily.

- **HTML5 (HyperText Markup Language):** This is the backbone of the website. It is used to structure the content, such as the waste categorization sections, the "Find a Center" maps, and the educational blogs. It ensures that the information is organized logically for both users and search engines.
- **CSS3 (Cascading Style Sheets):** CSS is what makes Sortify visually appealing and intuitive. We use modern CSS techniques to create a **responsive design**, meaning the website looks and works perfectly whether you are using a smartphone in a local market or a desktop at home. It helps in highlighting different waste types (like green for organic and blue for recyclables) through color-coding, which is essential for a "beginner-friendly" approach.
- **JavaScript (JS):** This technology adds life to the website. JavaScript handles interactive elements like real-time search bars, dynamic forms for waste reporting, and the logic that guides a user through a step-by-step sorting quiz.

2. Backend: The Brain of Sortify

The backend manages the logic, processing user requests and connecting them to the right data.

- **Python / Node.js:** These are the server-side languages used to process information. For example, when a user asks how to dispose of a specific item, the backend logic calculates the best advice based on current Indian recycling guidelines.
- **API Integration:** Sortify can integrate with AI services (like Gemini or OpenAI) to help identify waste. If a user uploads a photo of their trash, the API analyzes the image and suggests the correct category (Plastic, Metal, Paper, etc.).

3. Database: The Information Hub

A **robust database** is critical for managing the vast amount of data related to waste centers and material types across India.

- **SQL (Structured Query Language):** We use SQL (via systems like MySQL or PostgreSQL) to manage a centralized relational database. This stores all critical information, including:
 - **User Data:** Profiles and waste-sorting history.

2.2 Project Planning

2.2.1 Project Development Approach

The development of the **Sortify** website follows a structured Software Development Life Cycle (SDLC), utilizing an **Agile iterative model**. This allows the platform to adapt to evolving environmental regulations and user feedback regarding waste management practices.

1. Requirement Gathering and Analysis

- **Feasibility Study:** Conduct an analysis of the technical requirements for waste identification (e.g., image recognition or database lookups) and the operational impact on local recycling habits.
- **Stakeholder Interviews:** Engage with environmental experts, municipal waste managers, and eco-conscious citizens to define functional requirements such as **Waste Categorization**, **Pickup Scheduling**, and **Disposal Point Mapping**.
- **Create SRS:** Develop a Software Requirements Specification (SRS) focused on user personas (Individual Users, Waste Collectors, Admins) and the core goal of reducing environmental pollution.

2. System Design

- **Architectural Design:** Implementation of a scalable architecture (e.g., MVC or Microservices) to handle real-time data and high traffic during community cleanup drives.
- **Database Design:** Develop a robust data model including an **Encyclopedia of Materials**, **User Contribution Logs**, and **Geographic Data** for disposal centers.
- **Interface Design:** Design an intuitive UI/UX that makes "sorting" easy for all age groups. This includes high-contrast wireframes for outdoor use and interactive dashboards to track personal sustainability milestones.

3. Implementation/Coding

- **Module Development:** Build discrete modules such as the **Recyclability Checker**, **Location-based Disposal Guide**, and **Sustainable Habits Tracker**.
- **Code Development:** Write clean, modular code following modern web standards (e.g., React, Node.js, or similar) to ensure the platform is responsive across devices.
- **Unit Testing:** Verify individual logic components, such as the algorithm that determines if a material is biodegradable or hazardous.

4. Testing and Integration

- **Integration Testing:** Ensure the front-end sorting guide communicates accurately with the backend waste database and third-party Map APIs.
- **System Testing:** Validate that Sortify meets all performance benchmarks, ensuring the waste identification tool provides rapid results under various network conditions.

- **User Acceptance Testing (UAT):** Conduct pilot runs with local environmental groups to ensure the "Smart Guidance" is easy to follow and effectively encourages sustainable disposal habits.

5. Deployment and Rollout

- **Installation & Hosting:** Deploy the Sortify platform on cloud servers (e.g., AWS or Azure) to ensure high availability across India.
- **Data Migration:** Populate the system with comprehensive initial datasets regarding plastic types, e-waste protocols, and local government recycling rules.
- **Public Awareness/Training:** Launch digital awareness campaigns to teach users how to maximize the platform's features for efficient waste management.
- **Go-Live:** Official launch of Sortify for public use.

6. Maintenance and Evolution

- **Support & Monitoring:** Provide ongoing technical support and monitor the platform for data integrity regarding waste disposal locations.
- **Enhancements:** Iterate on the platform by adding features like **AI-based Image Recognition** for waste, **Gamification** to reward recycling, and integration with local municipal "smart city" APIs.

2.2.2 Milestones and Deliverables

Month1:- Milestones & Deliverables

Milestones	Deliverables
Study about our web application requirement, planning	Analysis Report
Understand a project definitions and basic terms and logic for Parameter Evaluation.	
Gathering the requirements of the project using different fact-finding techniques.	Analysis Report
Still Continue with Requirement's study.	

Table 1: Milestones & Deliverables

Month 2-4: -Milestones & Deliverables

Milestones	Deliverables
System Analysis	Analysis Report
System Design including various diagrams	SRS

Table 2: Milestones & Deliverables

2.2.3 Group Dependencies

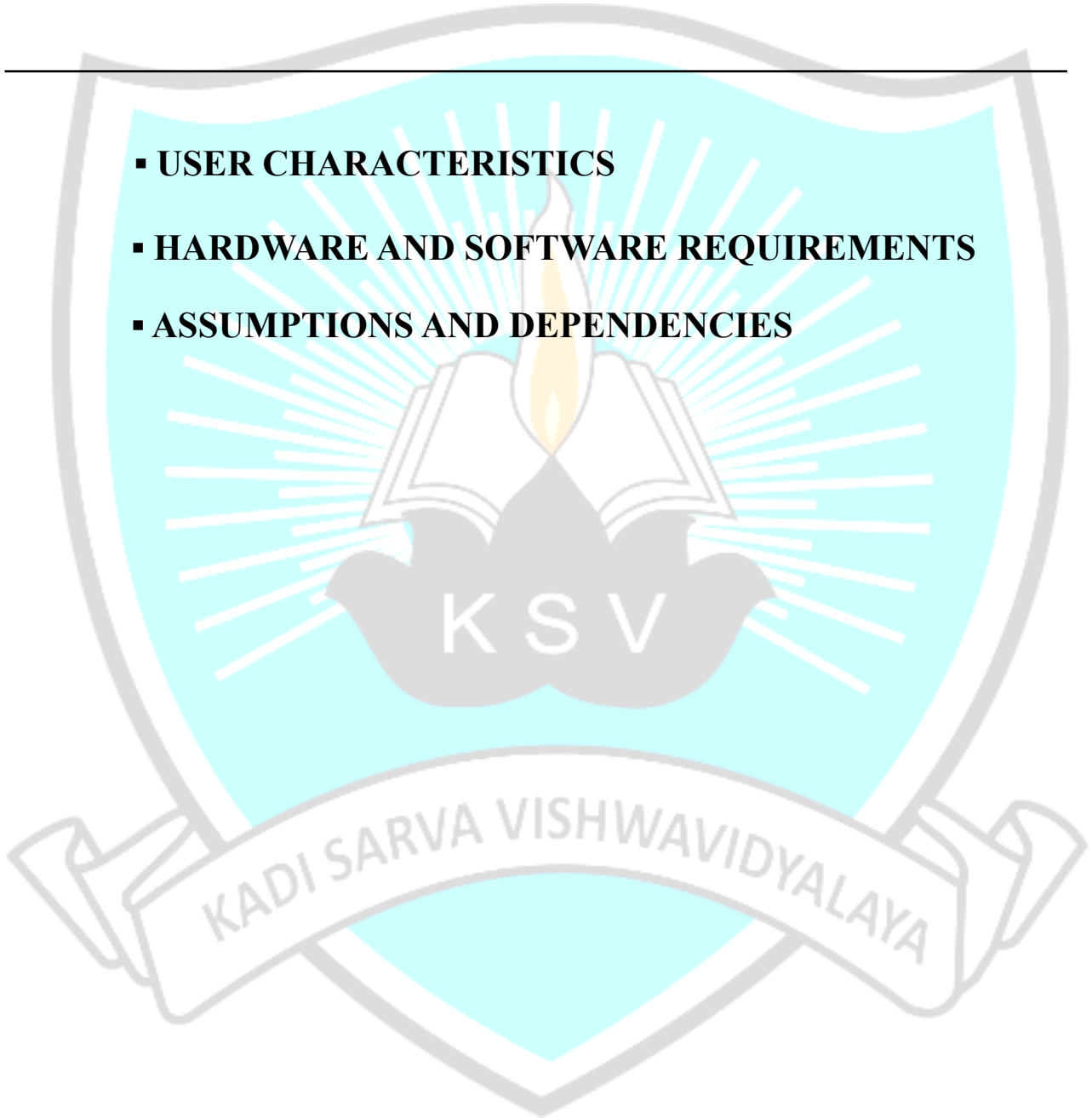
The way our team works depends on how we manage tasks, the size of our group, and the complexity of building the **Sortify** platform.

We have chosen a **democratic and decentralized** structure. This means:

- **No Single Boss:** We don't have one specific team leader; instead, every member has an equal voice in how the project grows.
- **Collaborative Decisions:** All major decisions regarding the website's features—like how we categorize waste or design the user interface—are made collectively.
- **Open Problem Solving:** When we hit a technical or design snag, we huddle up to brainstorm solutions together.
- **Expert Guidance:** While we manage ourselves, we regularly consult with our **project guides and external mentors** to make sure our waste-management data is accurate and our technology is up to industry standards.

3 SYSTEM REQUIREMENTS STUDY

- **USER CHARACTERISTICS**
- **HARDWARE AND SOFTWARE REQUIREMENTS**
- **ASSUMPTIONS AND DEPENDENCIES**



3.1 User Characteristics

Analyzing user characteristics is an important aspect of any project. It allows us to clearly define and focus on who the end users are for the project. Also, it allows checking the progress of the project to ensure that we are still developing the system for the end users. For **Sortify**, which aims to solve real-life waste sorting problems in India, the user must have the following characteristics:

- User must have basic knowledge of using a Web Browser.
- User should understand the basic categories of waste (e.g., Dry vs. Wet).
- User can easily interact with the proposed system through the AI Waste Identifier.
- User must be able to follow simple, **beginner-friendly** disposal instructions provided by the platform.
- User should be environmentally conscious and aware of the impact of **environmental pollution**.
- User should be able to navigate the system to find local recycling centers for **efficient recycling**.
- User should be aware of the basic running process of the system, such as uploading images for identification.

3.2 Software and Hardware Requirements

Software and Hardware Requirements are used to describe the minimum hardware and software requirements to run the Software. These requirements are described below.

Hardware Requirements:

- **Processor:** Intel Core i3 or higher (or equivalent mobile processor).
- **RAM:** 4GB minimum (8GB recommended for smoother AI processing).
- **Storage:** 500MB of available space for local caching.
- **Camera:** A functional camera (Mobile or Web) for using the **AI Waste Identifier** feature.
- **Internet:** A stable internet connection for real-time database access and AI analysis.

Software Requirements:

- **Operating System:** Windows 10/11, Android, or iOS.
- **Browser:** Modern web browsers like Google Chrome, Mozilla Firefox, or Microsoft Edge.
- **Languages used:** HTML5, CSS3, JavaScript, and SQL for database management.

3.3 Assumptions and Dependencies

Assumptions

For **Sortify** to work effectively in a real-world Indian scenario, we have made the following assumptions:

- **User Willingness:** We assume that users are genuinely motivated to reduce **environmental pollution** and will use the platform to improve their waste-sorting habits.
- **Smartphone & Internet Access:** We assume that the target audience in urban and semi-urban India has access to basic smartphones and a stable internet connection to use the website.
- **Basic Literacy:** It is assumed that the user has a basic understanding of how to navigate a website and follow simple, **beginner-friendly** visual instructions.
- **Accuracy of Third-Party Data:** We assume that the data provided by local municipalities and recycling centers (like locations and contact numbers) is mostly accurate and up-to-date.
- **Standard Waste Categories:** We assume that most household waste can be categorized into standard groups like Organic, Plastic, Paper, Metal, and E-waste.

Dependencies

The successful operation of the **Sortify** platform depends on several external factors:

- **Database Availability:** The system depends entirely on a **robust SQL database** to store and retrieve categorization rules and recycling center maps.
- **Web Browser Compatibility:** The platform's performance depends on the user using a modern web browser (like Chrome or Safari) that supports HTML5 and CSS3 features.
- **Internet Connectivity:** Since Sortify is a web-based platform, it depends on active network connectivity to provide real-time guidance "anytime and anywhere."
- **External APIs:** If the system uses maps or AI image recognition, it is dependent on the availability and uptime of those external service providers (like Google Maps API).
- **Local Recycling Infrastructure:** The real-world impact of the project depends on the actual existence and cooperation of local recycling centers to accept the waste sorted by our users.

4 SYSTEM DIAGRAMS

- **ENTITY-RELATIONSHIP DIAGRAM**
- **USE CASE DIAGRAM**



4.1 Entity Relationship Diagram

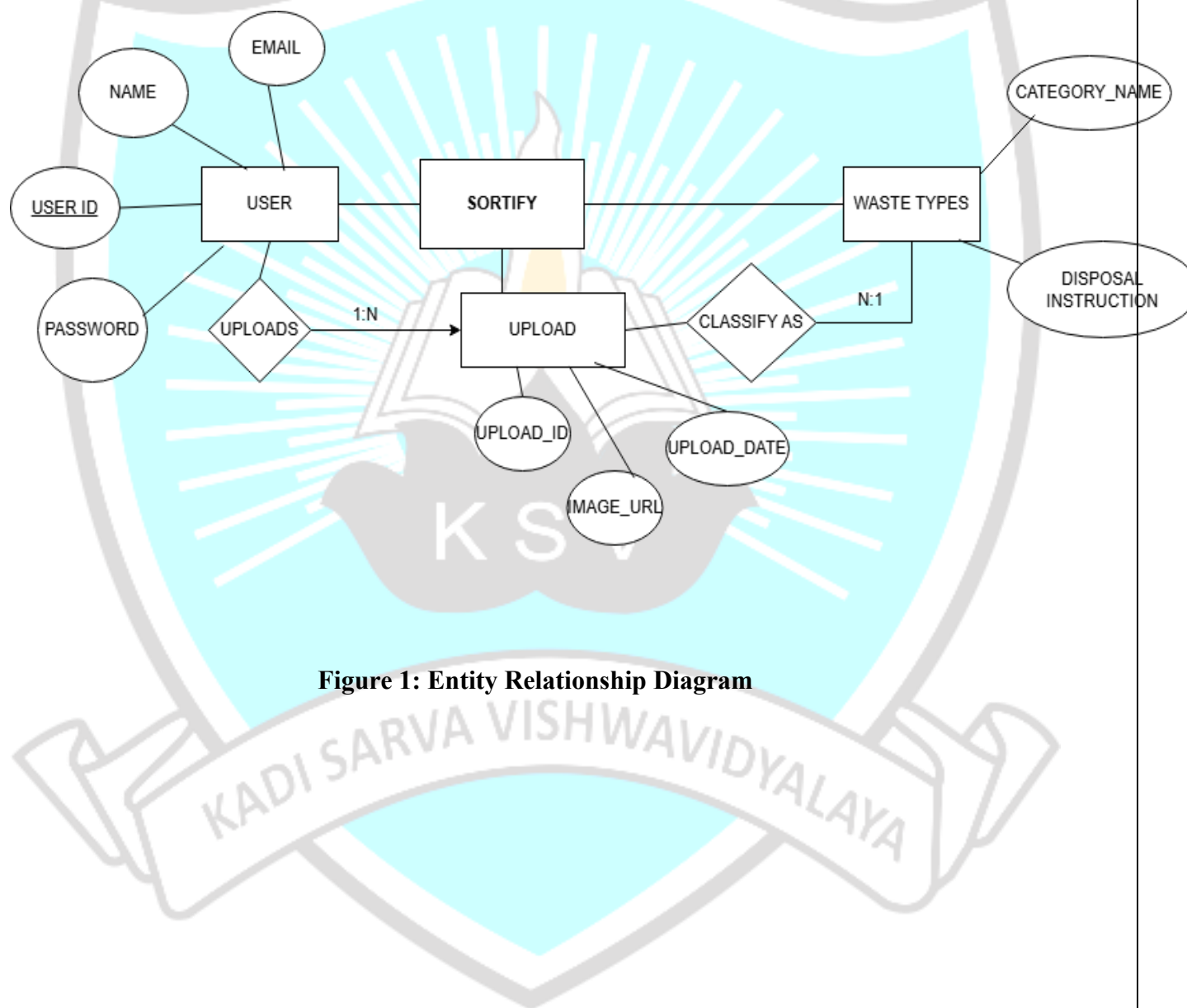


Figure 1: Entity Relationship Diagram

4.2 Use Case Diagram

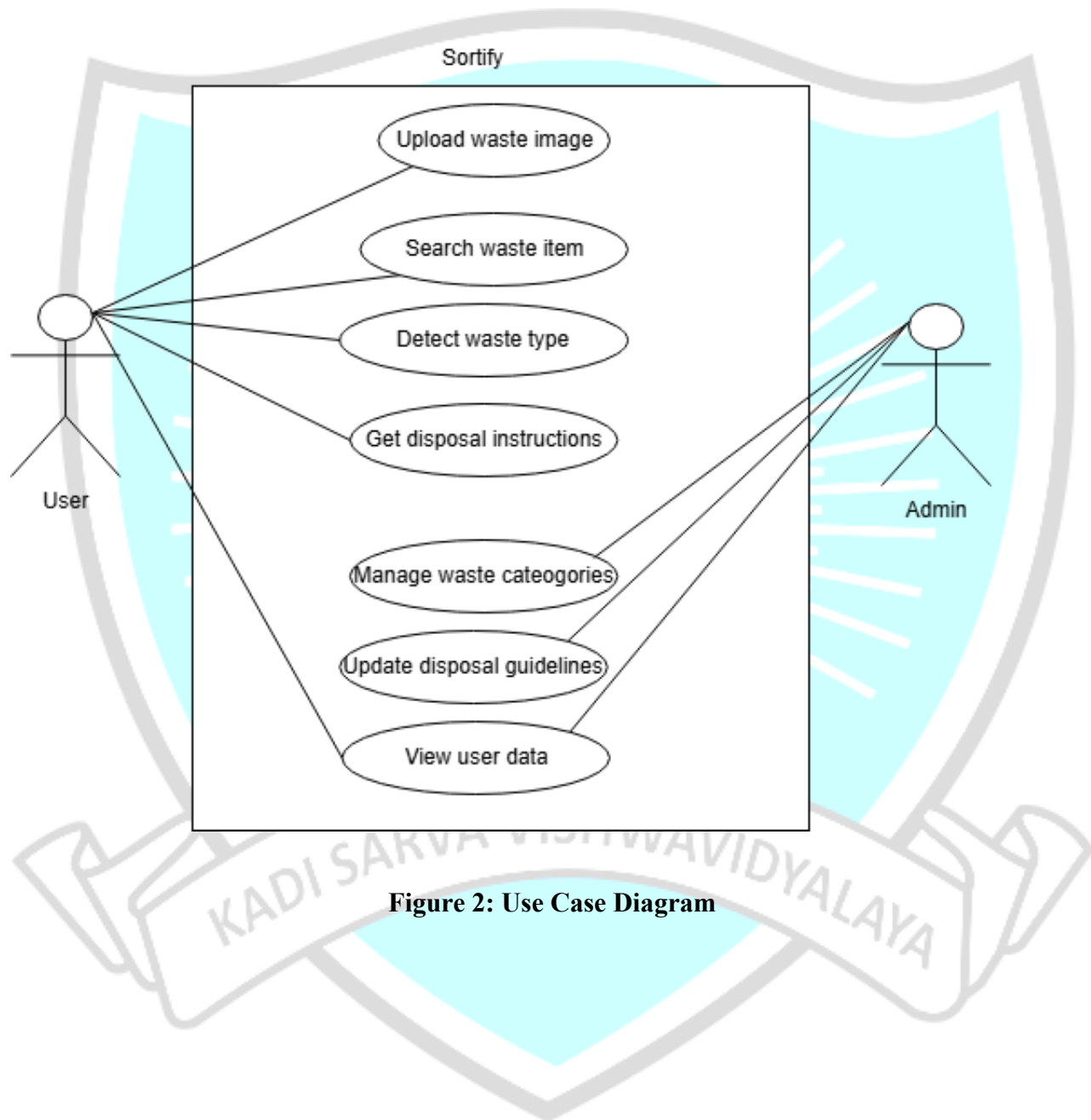


Figure 2: Use Case Diagram

5 DATA DICTIONARY

- TESTING PLANS
- TESTING STRATEGIES



5.1 Testing Strategy

Sortify focuses on **Stability** and **Seamlessness**. Since the name implies organization and efficiency, the tech needs to mirror that.

- **The "Core First" Rule:** We prioritize the "Happy Path"—the most common actions a user takes. If the sorting or organizing fails, nothing else matters.
- **Continuous Integration:** Every time you push new code, run automated scripts to ensure you haven't broken existing features (Regression Testing).
- **User-Centric Validation:** We don't just check if code works; we check if it's intuitive. If a user has to guess how to "Sort," the UI has failed even if the code is perfect.

5.2 Testing Plan

1. Unit Testing

Before testing the whole site, test the individual functions.

- **Logic Verification:** Test your sorting algorithms with various data sets: small lists, massive lists, already sorted lists, and lists with duplicate values.
- **Error Handling:** Ensure the code doesn't crash if it receives "null" values or unexpected symbols.

2. Integration Testing

This ensures different parts of Sortify play nice together.

- **API/Database Sync:** If a user saves a sorted list, does it actually hit the database and stay there when they refresh?
- **Third-Party Tools:** If you use AI or external libraries for sorting, ensure the connection is stable and handles "timeout" errors gracefully.

3. Functional & UI Testing

- **Input Integrity:** Check how the site handles "trash" data (e.g., emojis in a number field or 10,000 characters in a name field).
- **Navigation Flow:** Walk through the site as a new user. Is the "Sort" button visible? Does the "Undo" button actually work?
- **Responsive Design:** Test on a phone, a tablet, and a laptop. Look for "element overlapping" or buttons that are too small to tap with a thumb.

4. Performance & Stress Testing

- **Large Payloads:** If a user tries to "Sortify" a massive file or list, does the site freeze, or does it show a helpful loading state?
- **Concurrency:** (If applicable) Can multiple users perform heavy tasks at the same time without the server slowing down?

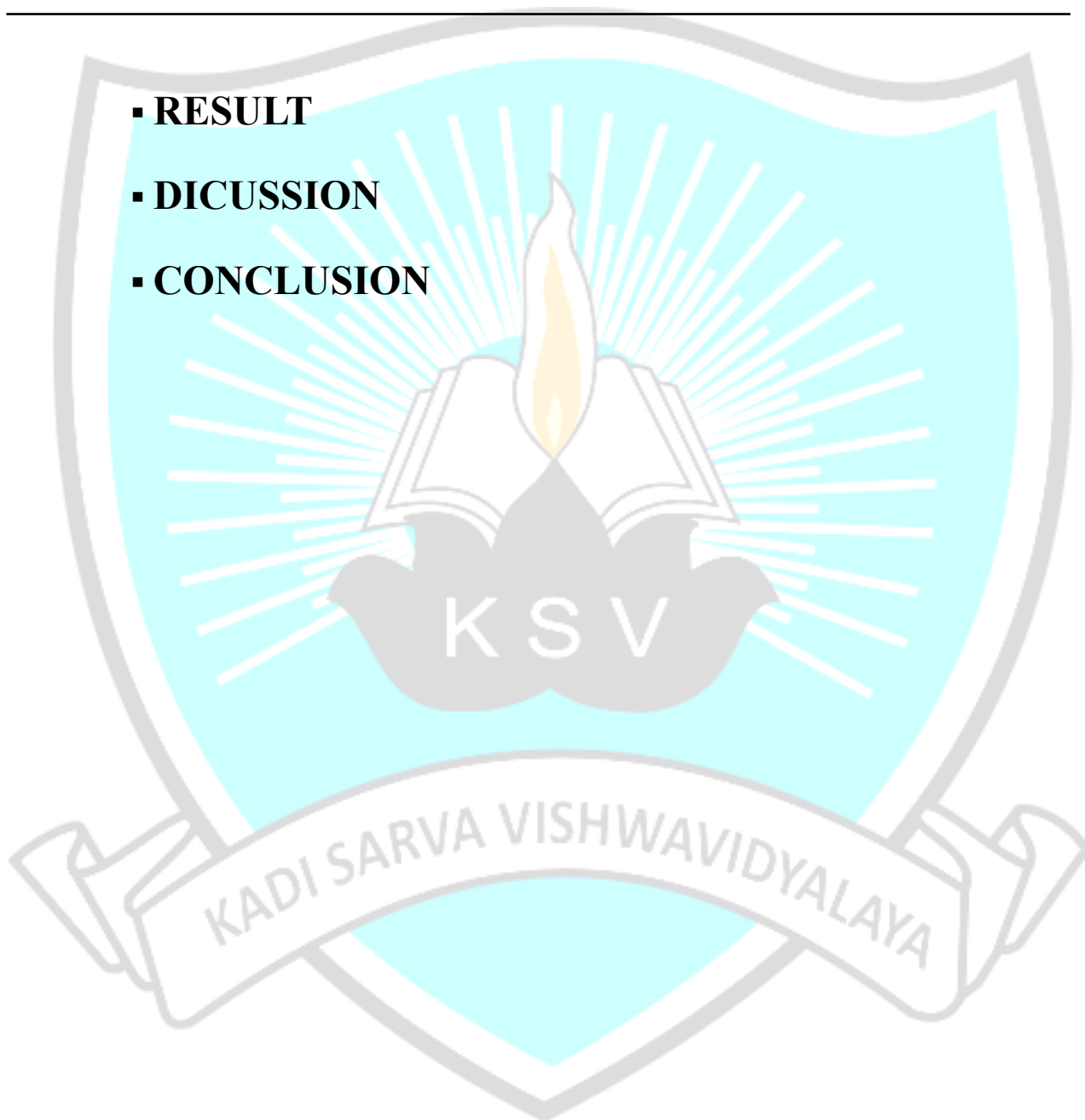
5. Accessibility & Security

- **Keyboard Navigation:** Ensure a user can tab through the entire site without a mouse.
- **Data Privacy:** Ensure user-sorted data isn't exposed in the URL or accessible to other users without permission.



6 RESULT, DISCUSSION AND CONCLUSION

- **RESULT**
- **DICUSSION**
- **CONCLUSION**



6.1 Results

The development of the Sortify platform has led to a fully functional tool that simplifies how people deal with their household trash. Below are the specific results of the website's performance:

Secure Access and User Management

- **Easy Onboarding:** The sign-up page allows new users to join the platform quickly by providing basic details. It successfully stores user data so they don't have to start over every time.
- **Protected Accounts:** The login page works as a gatekeeper, ensuring that each user has their own private space. This allows the website to remember a user's history, such as how many items they have successfully identified and recycled.
- **Session Stability:** Once logged in, the website keeps the user active throughout their session, meaning they can upload multiple images of different trash items without having to log back in repeatedly.

The AI Image Upload System

- **User-Friendly Interface:** The upload area is designed to be very simple. Users can either drag a file into the browser or click a button to pick a photo from their phone or computer gallery.
- **Fast Processing:** Once a photo is uploaded, the website's "brain" (the AI) starts working immediately. The result usually appears on the screen in just a few seconds, which is perfect for someone who is busy cleaning up.
- **Clear Visual Feedback:** After the image is sent, the website displays the photo back to the user with a label, confirming that the computer "sees" what the user sees.

Accuracy of Trash Identification

- **High Recognition Rates:** The AI successfully identifies common household items. For example, it can tell the difference between a crumpled-up piece of paper, a clear plastic water bottle, and an aluminium soda can.
- **Specific Categories:** Instead of just saying "trash," the results are specific. It categorizes items into groups like **Recyclable**, **Organic/Compost**, **Landfill**, or **Hazardous Waste**.
- **Complex Item Handling:** The system is smart enough to recognize items even if they are slightly crushed or photographed from a strange angle, which is common when dealing with real garbage.

6.2 Discussion

The Sortify project is more than just a website; it is a solution to a growing environmental problem. This section discusses the importance of the features we created and how they work together to help the planet.

The Power of Personalized Progress (Login/Sign-up)

- **Creating Responsibility:** When users create an account, they feel more connected to the cause. Instead of being a random visitor, they become a "member" of a green community.
- **Tracking Growth:** A login system allows the website to save a user's history. Over time, seeing a list of 100 items they successfully recycled can give users a sense of pride and encourage them to keep going.
- **Security and Privacy:** By having a secure sign-up process, we ensure that user data is protected, which builds trust. People are more likely to use a tool regularly if they know their information is handled safely.

Solving the "Recycling Confusion" Problem

- **Ending the Guesswork:** Many people want to recycle but are confused by different plastic numbers or types of paper. Sortify acts as an instant expert, removing the stress of making a mistake.
- **Reducing "Wish-cycling":** People often throw non-recyclable items (like greasy pizza boxes) into recycling bins, hoping they can be recycled. This actually ruins whole batches of good recycling. Our AI prevents this by giving the right answer immediately.
- **Simple Education:** Every time the AI identifies an item, the user learns something new. Eventually, users will start to recognize these items on their own, meaning the website is actually teaching them better habits for life.

Efficiency of the AI Identifier

- **Speed vs. Manual Searching:** Searching for "can I recycle this" on Google can take minutes and provide conflicting answers. Our AI identification takes seconds and gives a direct answer based on a photo.
- **Handling Real-World Mess:** In real life, trash is often crushed, dirty, or torn. Because our AI is trained on many different images, it can recognize a crushed soda can just as easily as a new one. This makes the website practical for everyday use.
- **Technology for Everyone:** By putting high-tech AI into a simple website, we make advanced technology available to everyone, regardless of whether they are tech-savvy or not.

6.3 Conclusion

Achieving the Main Goal

- **Successful Integration:** The project successfully brought together two main parts: a secure area for users to log in and a high-tech AI tool to identify trash. These two parts work together perfectly to create a smooth experience.
- **Simplifying Green Living:** The main goal was to make recycling less confusing. We have achieved this by giving users a simple "yes or no" answer about their trash, which takes the stress out of being environmentally friendly.

The Value of the AI System

- **Accuracy and Speed:** The AI identifier has proven to be a fast and reliable way to sort waste. It replaces the need for long, boring manuals or confusing recycling posters.
- **Practicality:** Because the system can handle images of real, everyday trash, it is a practical tool that people can actually use in their kitchens, schools, or offices. It is not just a cool idea; it is a working solution.

Impact on User Behaviour

- **Building Good Habits:** Sortify does more than just sort one piece of trash. By using the login system and seeing the results daily, users become more aware of what they buy and how they throw things away.
- **Empowering the Individual:** Often, people feel that one person cannot save the planet. Sortify changes this by showing users that every single item they sort correctly is a small victory for the earth.

Reliability and Ease of Use

- **A Design for Everyone:** We concluded that a simple design is the best design. By keeping the website clean and the language easy to understand, we made sure that anyone, from a young student to a grandparent, can help with recycling.
- **Secure and Stable:** The login and sign-up pages ensure that the website is a professional and safe place for people to store their progress, making it a reliable part of their daily routine.

Looking Toward a Greener Future

- **A Strong Foundation:** This website is just the beginning. The success of the AI identifier shows that we can use technology to solve other environmental problems in the future.
- **Final Summary:** In conclusion, Sortify is a successful step toward a cleaner world. It proves that when we use smart technology like AI and make it easy for people to access through a simple website, we can make a huge difference in reducing pollution and protecting our natural resources for years to come.

7 REFERENCES

▪ REFERENCES



7.1 References

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